

Introduction to research

***Research**

-research methodology is the systematic, logical, and scientific study of the methods applied to research.

-The first step is selecting a research problem/question.

-Studying previous research, books, reports, and articles related to the topic.

we can say Research is the process of gathering information and data to discover a new knowledge/concept or advancement of existing theories, which is a new understanding that was not previously known.

-*Data Collection

Two major types:

•Primary Data → Collected first-hand (survey, interviews, observations, experiments).

•Secondary Data → Already available (books, reports, journals, websites, government data).

-*Data Analysis

Organizing and interpreting the data.

Using statistical tools (Excel, SPSS, R, Python) to analyze trends and relationships.

-Ensures research is systematic and scientific.

Provides accuracy and reliability in findings.

Helps in decision making for organizations, industries, and governments.

Leads to the creation of new knowledge and solutions.

***Objective of research**

- objective of research in research methodology is to systematically investigate and understand a phenomenon, problem, or question, ultimately contributing to the expansion of knowledge or the development of new solutions.

-1. To Gain Knowledge & Understanding

The primary aim is to discover new facts, ideas, and insights.

Example: Studying how artificial intelligence affects education.

2. To Describe Phenomena:

Research helps in giving a clear picture of a situation or problem.

Example: Describing the level of pollution in a particular city.

4. To Test Hypotheses

5. To Predict Outcomes:

Based on data, research can forecast future trends.

6. To Find Solutions to Problems (Applied Research)

Research provides practical solutions to social, economic, medical, and technical problems.

Example: Developing a new medicine for a disease.

Motivation in research

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Types of research

-1. Qualitative Research

Qualitative research refers to analyzing in-depth information about human behavior and producing “textual data” (non-numerical).

Meaning: Focuses on understanding experiences, meanings, and concepts rather than numbers.

Data: Non-numerical (words, images, videos, observations).

Example: Studying how teachers feel about online learning.

2. Quantitative Research

Quantitative research refers to analyzing something based on some numerical data and mathematical models and producing “numerical data”.

Meaning: Focuses on measurable data and uses statistical analysis.

Data: Numerical (percentages, scores, frequencies).

Example: Measuring exam scores of students before and after online learning.

3. Mixed Methods Research

Meaning: Combines both qualitative and quantitative approaches to get a more complete understanding.

Data: Both numerical and descriptive.

These combine both quantitative and qualitative approaches to provide a more comprehensive understanding of the research topic.

-Qualitative = Words, meanings, opinions

Quantitative = Numbers, measurements, statistics

Mixed Methods = Both together

Research approaches

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1. Qualitative approach

2 quantitative approach

3. Mixed methods approach

Accountancy

Business Studies

Economics

Organisational Behaviour

Human Resource Management

Entrepreneurship

Marketing

Income Tax

Finance

Management

Commerce



Introduction to Research Methodology

The term Research may be defined as a systematic gathering or collection of data and information. It is a process of analysis for the advancement of knowledge in any subject.

Definition of Research

There is various definition of Research by various researchers/ as per the fields of their study and the availability of resources at the given time. As

a basic definition of Research, we can say Research is the process of gathering information and data to discover a new knowledge/concept or advancement of existing theories, which is a new understanding that was not previously known.

Characteristics of good research

The purpose of the research is clear.

It follows a systematic process.

It is logical and objective.

It is easy to understand.

Research starts with a problem/question.

It creates a path for generating new questions.

It is ethical.

It resolves any current problems/issues.

The data of the research is appropriate.

It is empirical & replicable.

Research & characteristics of research

Research is a process of collecting, analyzing and interpreting information to answer questions.

But to qualify as research, the process must have certain characteristics: it must, as far as possible,

1. Systematic – Research follows a step-by-step process (problem → objectives → data collection → analysis → conclusion).
2. Logical – Based on reasoning and rational thinking.
3. Empirical – Depends on observation and real evidence (not assumptions).
4. Replicable – Other researchers should be able to repeat it and get similar results.
5. Objective – Free from personal bias or emotions.
6. Valid & Reliable – Findings must be accurate (valid) and consistent (reliable).
7. Critical – Examines facts carefully and questions assumptions.
8. Innovative – Should generate new knowledge, ideas, or methods.
9. Problem-Oriented – Aims to find solutions to specific problems.
10. Ethical – Conducted honestly, without manipulation or harm to participant

Ethics and integrity

Key Principles of Research Ethics

Honesty – Report data, results, and methods truthfully.

Integrity – Keep promises, follow rules, and maintain transparency.

Fairness – Avoid discrimination, give credit to contributors (no plagiarism).

Confidentiality – Keep participants' personal information private.

Respect for Participants – Protect rights, dignity, and privacy of research subjects.

2. Research Integrity

Honest reporting → No falsification, fabrication, or plagiarism.

Transparency → Clearly explain methods and procedures.

Reproducibility → Work should be reliable so others can verify results.

Proper authorship → Give credit to all contributors.

Plagiarism & types of plagiarism

- Plagiarism:

📌 Meaning:

-Plagiarism is the act of using someone else's words, ideas, data, or work without giving proper credit, and presenting it as your own.

-Plagiarism, in simple terms, is taking someone else's work or ideas and presenting them as your own, without giving them proper credit.

Types of Plagiarism

1. Direct (Verbatim) Plagiarism

- Copying text word-for-word without quotation marks or citation.
- ❌ Example: Copy-pasting from a website into an assignment.

2. Self-Plagiarism

- Reusing your own previously published work (or assignments) without citing it.
- ❌ Example: Submitting the same project in two different classes.

3. Mosaic (Patchwork) Plagiarism

Mixing copied phrases with your own words, or replacing some words with synonyms, without citation.

❌ Example: Copying parts of Wikipedia and changing a few words.

❌ Example: Using a sentence from a book but forgetting to mention the author.

6. Source-based Plagiarism

Incorrectly citing sources or using non-existent/false references.

✗ Example: Quoting a paper but writing the wrong author name.

7. Global (Complete) Plagiarism

Submitting someone else's entire work as your own.

✗ Example: Buying an assignment from the internet and submitting it.

Plagiarism check tools

Plagiarism check tools are software or online applications that help detect whether a piece of writing contains copied content from other sources.

They compare your text against a large database of books, journals, websites, and student papers to find similarities.

👉 In simple words: These tools check if your work is original or copied.

How Do They Work?

You upload or paste your text into the tool.

The tool scans the content and compares it with databases.

It highlights matched or similar text.

It gives a similarity score/percentage (e.g., 15% plagiarized).

It provides the sources/links from which the content was copied.

Examples of Plagiarism Check Tools:

Turnitin – widely used in universities.

Grammarly Plagiarism Checker – checks against online sources.

Quetext – simple tool for quick checks.

Research methods versus methodology

1. Research Methods

- **Meaning:** The specific techniques or tools used to collect and analyze data.
- **Focus:** **How** the research is conducted.
- **Nature:** Practical in nature.
- **Examples:**
 - Surveys
 - Interviews
 - Experiments
 - Observations
 - Questionnaires
 - Statistical analysis

👉 In simple words: **Methods = Tools & techniques.**

2. Research Methodology

- **Meaning:** The **overall strategy, philosophy, and rationale** behind the research. It explains *why* certain methods are chosen.
- **Focus:** The **logic behind the methods** and the approach used.
- **Nature:** Theoretical in nature.
- **Includes:**
 - Research design (qualitative, quantitative, mixed)
 - Justification for using certain methods
 - Limitations of chosen methods
 - Underlying assumptions and principles

👉 In simple words: **Methodology** = The science of methods (the “why”).

Research & scientific method

Steps of the Scientific Method in Research

-Identify the Problem / Question

Define what you want to study.

Example: Does social media affect students' academic performance?

-Review of Literature

Study existing work to understand what is already known.

-Form Hypothesis

A testable statement predicting the relationship between variables.

Example: “Excessive social media use negatively impacts students' grades.”

-Research Design & Methodology

Decide whether the study will be qualitative, quantitative, or mixed.

-Select tools: surveys, experiments, interviews, etc.

-Data Collection

Gather information systematically (e.g., questionnaires, lab experiments).

-Data Analysis

Use statistical tests (quantitative) or thematic coding (qualitative).

-Interpretation of Results

Compare results with the hypothesis.

Example: Survey shows students using >3 hrs/day on social media have lower grades.

-Conclusion & Recommendations

Confirm or reject hypothesis. Suggest future research or applications.

-Reporting & Publication

Share findings with the academic community (thesis, paper, journal).

Importance of knowing how research is done

Knowing how research is done is important because it makes knowledge reliable, improves problem-solving, develops critical thinking, ensures ethics, supports innovation, and benefits society

Understanding how research is conducted is essential for students, professionals, and society because it ensures that knowledge is reliable, accurate, and useful

📌 1. Builds Systematic Knowledge

Research follows a structured process (problem → data → analysis → conclusion).

Knowing this process helps in generating valid and reliable knowledge.

📌 2. Helps in Problem-Solving

Research teaches us how to identify a problem, collect evidence, and find solutions.

Example: A teacher can research better teaching methods to improve learning outcomes.

📌 3. Develops Critical Thinking

Understanding research methods helps to question assumptions, analyze facts, and avoid bias.

You don't just accept information—you evaluate its source and validity.

📌 4. Improves Decision-Making

Research provides evidence-based information.

Example: Governments use research to make policies on health, education, and economy.

📌 5. Prevents Plagiarism and Misconduct

Knowing research procedures ensures ethical practices, proper citation, and originality.

📌 6. Supports Academic & Professional Growth

In academics, research skills are vital for writing projects, dissertations, and publications.

In jobs, professionals use research to innovate and improve performance.

📌 8. Contributes to Society

Research in medicine, education, environment, etc. improves quality of life.

Example: COVID-19 vaccines were developed through proper research methods.

Criteria of good research

---Clear, systematic, empirical, logical, objective, reliable, valid, replicable, ethical, and contributes to knowledge.

1. Clarity of Purpose

- The **problem and objectives** of the research must be clearly defined.
- Example: Instead of a vague question like “Does social media affect students?”, a clear one would be “What is the impact of social media usage on the academic performance of college students?”
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2. Systematic approach

*Research should follow a step-by-step scientific method (problem → hypothesis → data → analysis → conclusion).
Random or unorganized work is not reliable.*

3. Empirical

Findings should be based on observations, experiments, or data, not on personal opinions.

4. Logical and Rational

The process must be based on reasoning, with conclusions supported by evidence.

5. Objectivity

*Research must be free from personal bias or emotions.
Data, not the researcher’s opinion, should guide the results.*

6. Replicability

Other researchers should be able to repeat the study under similar conditions and get similar results.

7. Reliability and Validity

Reliability → Results should be consistent if repeated.

Validity → Research must actually measure what it intends to measure.

8. Ethical Considerations

Must respect participants’ rights, consent, and confidentiality.

No falsification, fabrication, or plagiarism.

9. Precision and Accuracy

Data collection and analysis must be careful and exact to avoid errors.

10. Contribution to Knowledge

Good research should add something new, useful, or meaningful to existing knowledge.

It should also suggest future research directions.

✔ In short

Good research is:

Clear, systematic, empirical, logical, objective, reliable, valid, replicable, ethical, and contributes to knowledge.

Sampling:

This involves selecting a subset of the population to study, ensuring it's representative of the larger group.

Data Collection:

This includes choosing the appropriate tools and techniques for gathering data, such as questionnaires, interviews, observations, or experiments.

Data Analysis:

This step involves processing and interpreting the collected data, using statistical methods, thematic analysis, or other relevant techniques.

Ethical Considerations:

Researchers must adhere to ethical guidelines, ensuring the well-being and privacy of participants, obtaining informed consent, and maintaining confidentiality.

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